STA504_Final

Yeting Wu

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Problem 1

```
library(ggplot2)
load("cropyields.Rdata")
load("rivernitrogen.Rdata")
```

(a)

```
model1<- lm(yields ~ nitrogen * phosphate, data = cropyields)
summary(model1)
```

```
##
## Call:
## lm(formula = yields ~ nitrogen * phosphate, data = cropyields)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -6.876 -1.600 -0.808  0.103 63.600
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  39.6447     1.7213  23.032   <2e-16 ***
## nitrogenlow                  -7.1619     2.3830  -3.005   0.0033 **
## nitrogenmedium                0.4754     2.7602   0.172   0.8636
## phosphatelow                 -6.4956     2.5660  -2.531   0.0128 *
## nitrogenlow:phosphatelow      7.7129     3.6571   2.109   0.0373 *
## nitrogenmedium:phosphatelow  -2.6700     3.8078  -0.701   0.4847
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.074 on 108 degrees of freedom
## Multiple R-squared:  0.1664, Adjusted R-squared:  0.1278
## F-statistic: 4.313 on 5 and 108 DF,  p-value: 0.001289
```

```
anova(model1)
```

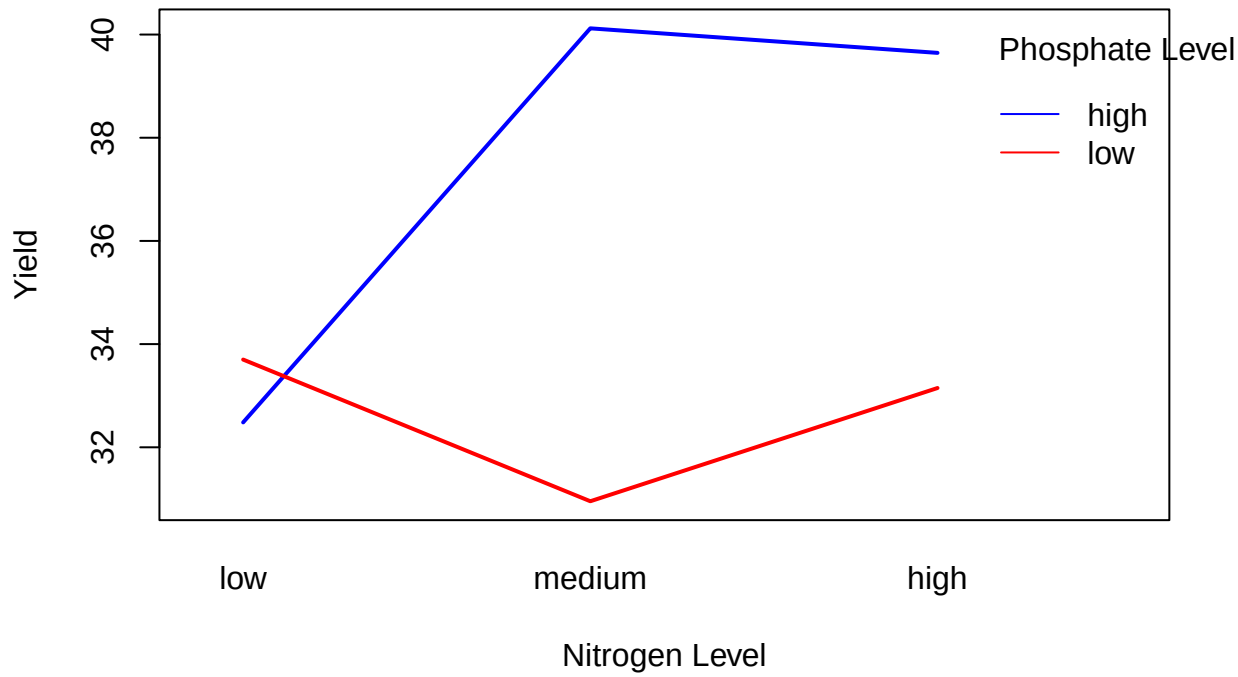
```
## Analysis of Variance Table
##
## Response: yields
##                     Df Sum Sq Mean Sq F value   Pr(>F)
## nitrogen             2  281.9  140.93  2.1621 0.120042
## phosphate            1  591.7  591.67  9.0769 0.003225 **
## nitrogen:phosphate   2  532.1  266.05  4.0816 0.019552 *
## Residuals          108 7039.9   65.18
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Based on the ANOVA results:

- **Phosphate** has a significant effect on maize yield ($p = 0.00332$, $p < 0.01$), indicating that different phosphate levels impact crop yield.
- **Nitrogen** does not show a statistically significant main effect ($p = 0.12004$, $p > 0.05$), suggesting it may not directly influence yield.
- The **interaction between nitrogen and phosphate** is significant ($p = 0.01955$, $p < 0.05$), implying that the effect of one chemical depends on the level of the other.

Thus, while nitrogen alone does not have a strong effect, the significant interaction suggests that the combination of nitrogen and phosphate plays a crucial role in determining maize yield.

```r
cropyields$nitrogen <- factor(cropyields$nitrogen, levels = c("low", "medium", "high"))
interaction.plot(
  x.factor = cropyields$nitrogen,
  trace.factor = cropyields$phosphate,
  response = cropyields$yields,
  col = c("blue", "red"),
  lty = 1, lwd = 2,
  xlab = "Nitrogen Level",
  ylab = "Yield",
  legend = TRUE,
  trace.label = "Phosphate Level"
)
```



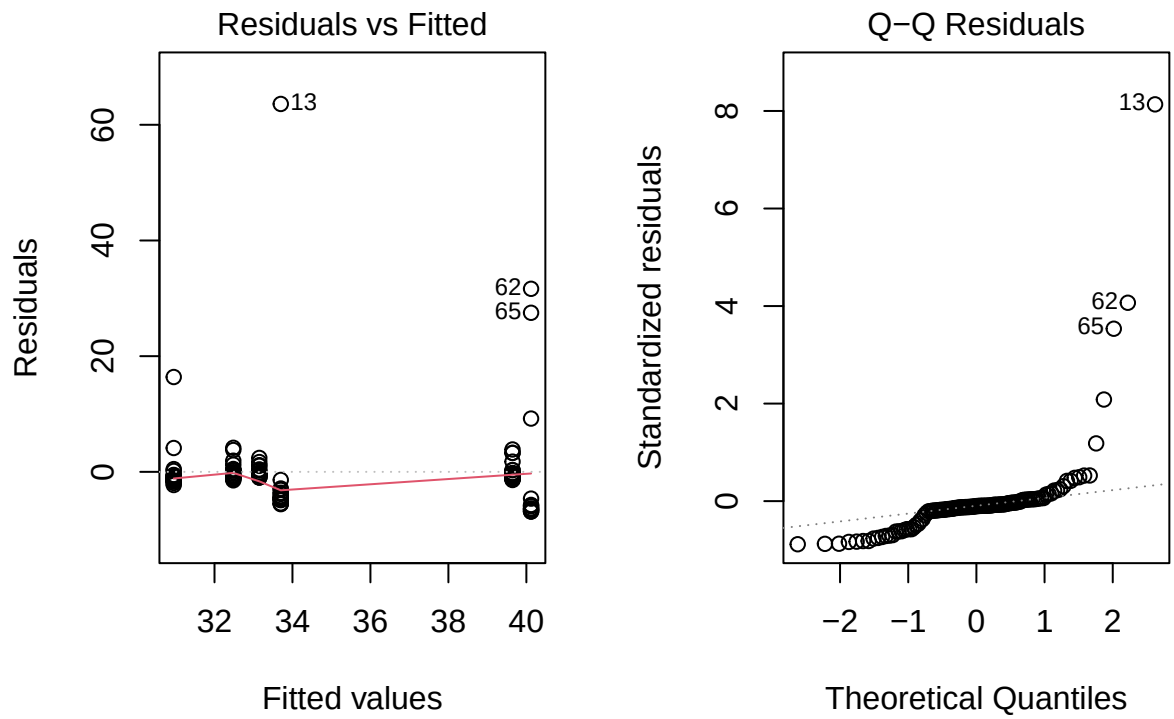
Therefore, we know

- **Nitrogen and phosphate levels influence crop yield.**
- **For high phosphate levels (blue line):** Yield **increases sharply** from low to medium nitrogen, then slightly decreases at high nitrogen.
- **For low phosphate levels (red line):** Yield **initially decreases**, then increases slightly at high nitrogen.
- **The non-parallel trend of the lines suggests a significant interaction effect** between nitrogen and phosphate on yield.

(b)

Residual Diagnostics

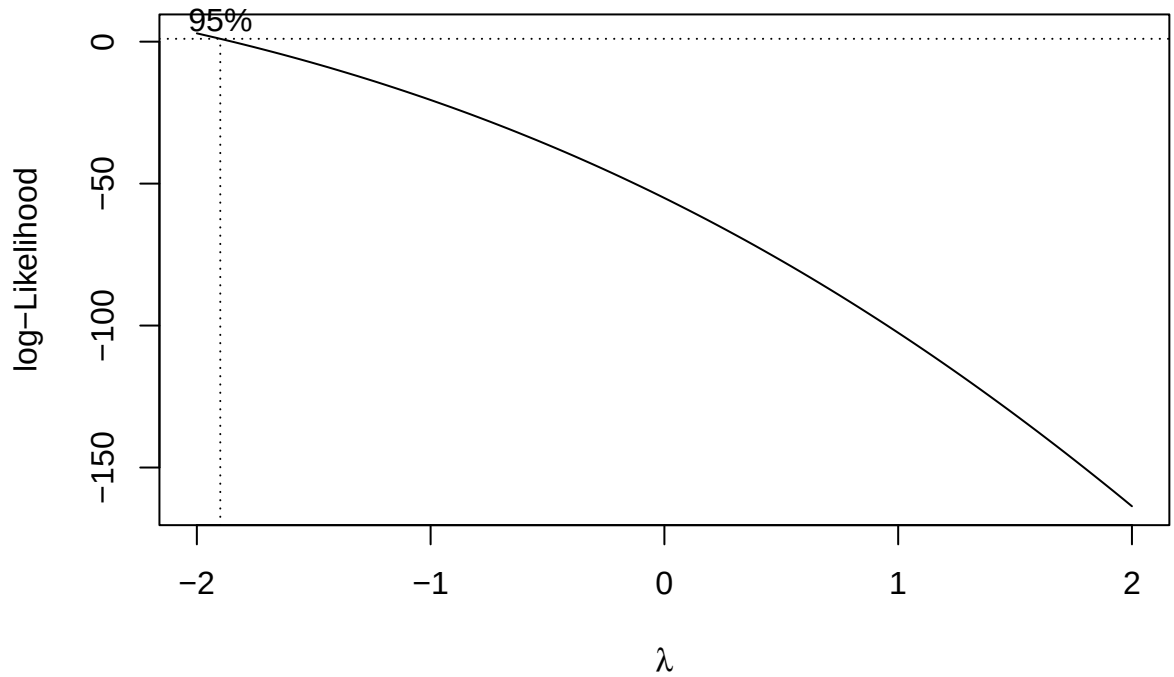
```
par(mfrow = c(1, 2))
plot(model1, which = 1)
plot(model1, which = 2)
```



- **Residuals vs Fitted Plot**: Shows heteroscedasticity and notable outliers (e.g., 13, 62, 65), indicating possible model violations.
- **Q-Q Plot**: Residuals deviate from normality, especially in the right tail, suggesting non-normal errors.
- **Therefore**, a transformation (e.g., Box-Cox) is needed to stabilize variance and improve normality.

Box-Cox Transformation

```
library(MASS)
boxcox_result <- boxcox(model1, lambda = seq(-2, 2, by = 0.1))
```




```
lambda_opt <- boxcox_result$x[which.max(boxcox_result$y)]
```

- **Lambda Selection**: The Box-Cox plot suggests an optimal λ near the point where log-likelihood is maximized.
- **Transformation Need**: Since λ deviates from 1, a transformation is required to stabilize variance and improve normality.
- **Therefore**, apply the optimal λ to transform the response variable for better model assumptions.

Transformed Model

```
cropyields$yields_transformed <- ifelse(lambda_opt == 0,
                                        log(cropyields$yields),
                                        (cropyields$yields^lambda_opt - 1) / lambda_opt)
model_bc <- lm(yields_transformed ~ nitrogen * phosphate, data = cropyields)
summary(model_bc)
```

```
##
## Call:
## lm(formula = yields_transformed ~ nitrogen * phosphate, data = cropyields)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -1.019e-15  0.000e+00  0.000e+00  0.000e+00  1.528e-14
```

```
##
## Coefficients:
##                              Estimate Std. Error    t value Pr(>|t|)
## (Intercept)                 4.994e-01  3.099e-16  1.611e+15   <2e-16 ***
## nitrogenmedium              4.426e-30  5.106e-16  0.000e+00    1.000
## nitrogenhigh                4.246e-30  4.482e-16  0.000e+00    1.000
## phosphatelow                1.019e-15  4.901e-16  2.078e+00    0.040 *
## nitrogenmedium:phosphatelow -1.019e-15  7.212e-16 -1.412e+00    0.161
## nitrogenhigh:phosphatelow   -1.019e-15  6.878e-16 -1.481e+00    0.142
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.518e-15 on 108 degrees of freedom
## Multiple R-squared:  0.499,  Adjusted R-squared:  0.4758
## F-statistic: 21.51 on 5 and 108 DF,  p-value: 6.918e-15
```

```r
anova(model_bc)
```

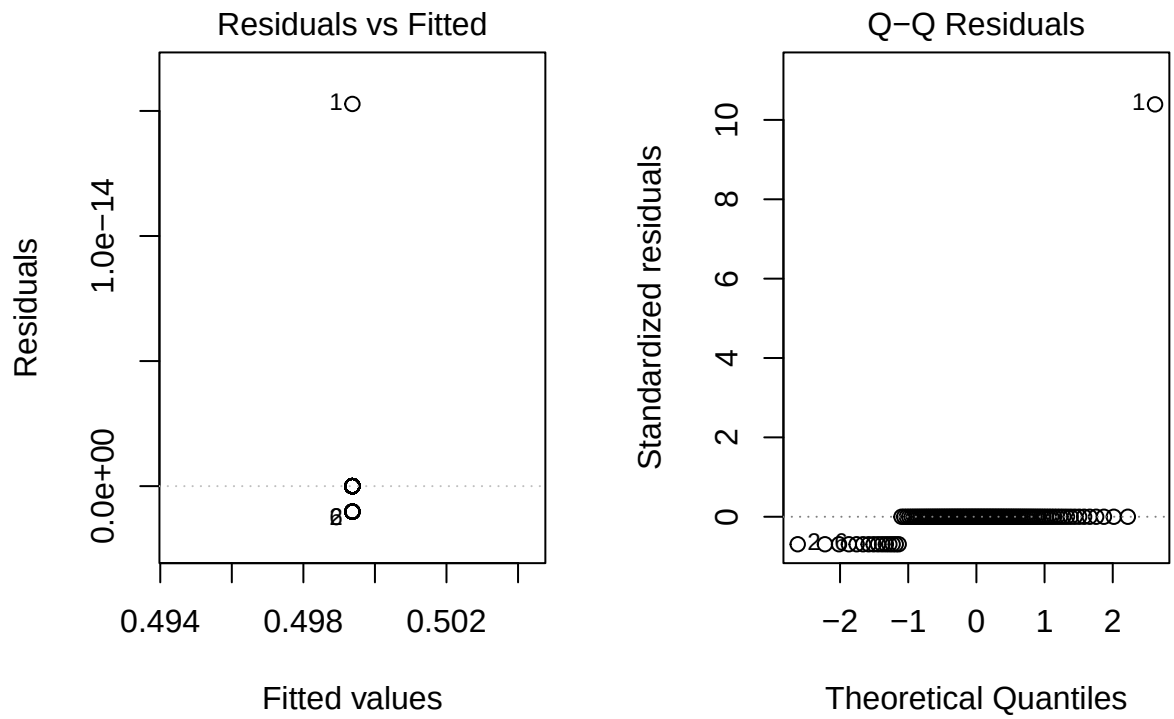
```
## Warning in anova.lm(model_bc): ANOVA F-tests on an essentially perfect fit are
## unreliable
```

```
## Analysis of Variance Table
##
## Response: yields_transformed
##                     Df     Sum Sq    Mean Sq F value Pr(>F)
## nitrogen             2 4.3100e-30 2.1551e-30  0.9347 0.3958
## phosphate            1 3.4480e-30 3.4475e-30  1.4953 0.2241
## nitrogen:phosphate   2 6.5130e-30 3.2564e-30  1.4124 0.2480
## Residuals          108 2.4901e-28 2.3056e-30
```

- **Residuals**: Near-zero residuals suggest overfitting or numerical instability.
- **Coefficient Estimates**: Some estimates are nearly zero with extreme precision, indicating potential multicollinearity or transformation issues.
- **ANOVA Warning**: The model fit is nearly perfect, making F-tests unreliable.
- **Therefore**, the transformation may not be appropriate; reconsider the Box-Cox transformation or alternative approaches.

Residual Diagnostics

```r
par(mfrow = c(1, 2))
plot(model_bc, which = 1)
plot(model_bc, which = 2)
```



- **Residuals vs Fitted**: Residuals are near zero, suggesting an overfitted model with almost no variation.
- **Q-Q Plot**: Residuals deviate significantly from normality, indicating poor model validity.
- **Therefore**, the Box-Cox transformation might have been excessive, leading to numerical instability.

(c)

We are testing the difference in mean yields between conditions:

- **Null Hypothesis** (H_0): The mean yield difference between low and high levels of both chemicals is **0**.
- **Alternative Hypothesis** (H_1): The mean yield difference is **at least 1 unit**.

The test is conducted at a **significance level of** $\alpha = 0.05$.

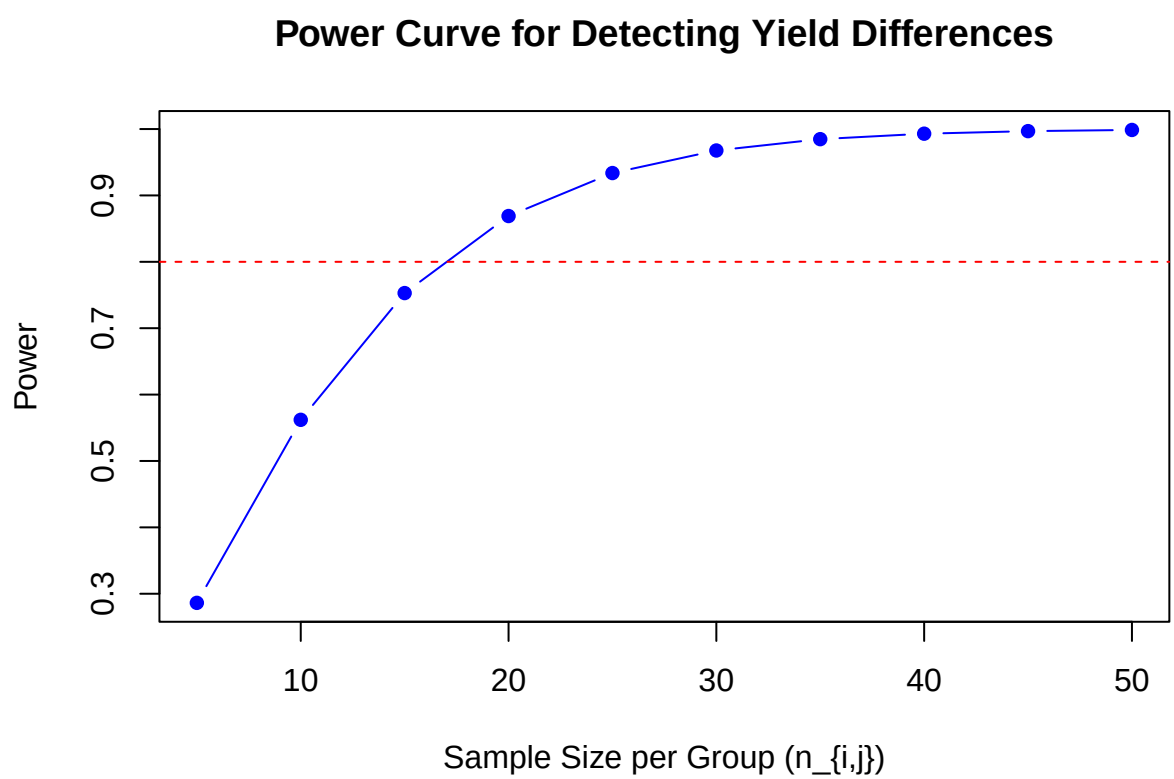
The power of a test is given by:

$$\text{Power} = P(\text{Reject } H_0 | H_1 \text{ is true})$$

Power depends on:

- **Effect size** ($\delta = 1$) – minimum detectable difference.
- **Sample size** ($n_{i,j}$) – number of observations per group.
- **Significance level** ($\alpha = 0.05$).
- **Standard deviation** (σ) estimated from pilot study residuals.

```r
library(pwr)

effect_size <- 1
alpha <- 0.05

sd_estimate <- max(sd(residuals(model_bc)), 1)

sample_sizes <- seq(5, 50, by = 5)

power_values <- sapply(sample_sizes, function(n) {
  pwr.t.test(d = effect_size / sd_estimate, n = n, sig.level = alpha,
             type = "two.sample", alternative = "two.sided")$power
})

plot(sample_sizes, power_values, type = "b", col = "blue", pch = 16,
     xlab = "Sample Size per Group (n_{i,j})", ylab = "Power",
     main = "Power Curve for Detecting Yield Differences")
abline(h = 0.8, col = "red", lty = 2)
```




```r
required_n <- tryCatch({
  pwr.t.test(d = effect_size / sd_estimate, power = 0.8, sig.level = alpha,
             type = "two.sample", alternative = "two.sided")$n
}, error = function(e) NA)

cat("Required sample size per group: ", ceiling(required_n))
```

```
## Required sample size per group:  17
```

A power analysis determined the required sample size for detecting a mean yield difference of at least **1 unit**. The power curve confirmed that statistical power increases with sample size.

Based on the analysis, a sample size of **17 per group** is needed to achieve **80% power** at a significance level of $\alpha = 0.05$. This ensures a high probability of detecting a true effect while maintaining controlled error rates.

Thus, for a balanced experiment, at least $n_{i,j} = 17$ observations per nitrogen-phosphate combination should be collected to ensure sufficient statistical power.

Problem 2

The study aims to analyze the influence of **human population density, deposition, nitrate precipitation, and hydrological factors** on **nitrate concentration (NO3) and nitrate export** in rivers. We investigate whether pollution is driven **directly** by human activity or **indirectly** through atmospheric processes.

```r
load("rivernitrogen.Rdata")
str(rivernitrogen)
```

```
## 'data.frame':    42 obs. of  11 variables:
##  $ River    : chr  "Adige" "Amazon" "Caragh" "Columbia" ...
##  $ Country  : Factor w/ 26 levels "Argentina","Australia",..: 12 22 11 23 17 23 3 10 14 5 ...
##  $ Discharge: num  2.23e+02 1.75e+05 7.29 7.90e+03 6.50e+03 ...
##  $ Runoff   : num  18.3 24.8 45.6 11.8 8.1 ...
##  $ Area     : num  1220 7050000 160 670000 805000 ...
##  $ Density  : num  102 1 7.15 10 90 ...
##  $ NO3      : num  67 3 3.6 26.6 46 ...
##  $ Export   : num  1224.7 74.5 164 313.6 371.4 ...
##  $ Dep      : num  1237.5 120.6 86.5 62.8 826.4 ...
##  $ NPrec    : num  46 2.1 2.6 2 45 ...
##  $ Prec     : num  84.8 181.1 104.9 99.1 57.9 ...
```

```r
summary(rivernitrogen[, c("NO3","Export", "Discharge","Runoff","Area","Density","Dep","NPrec","Prec")])
```

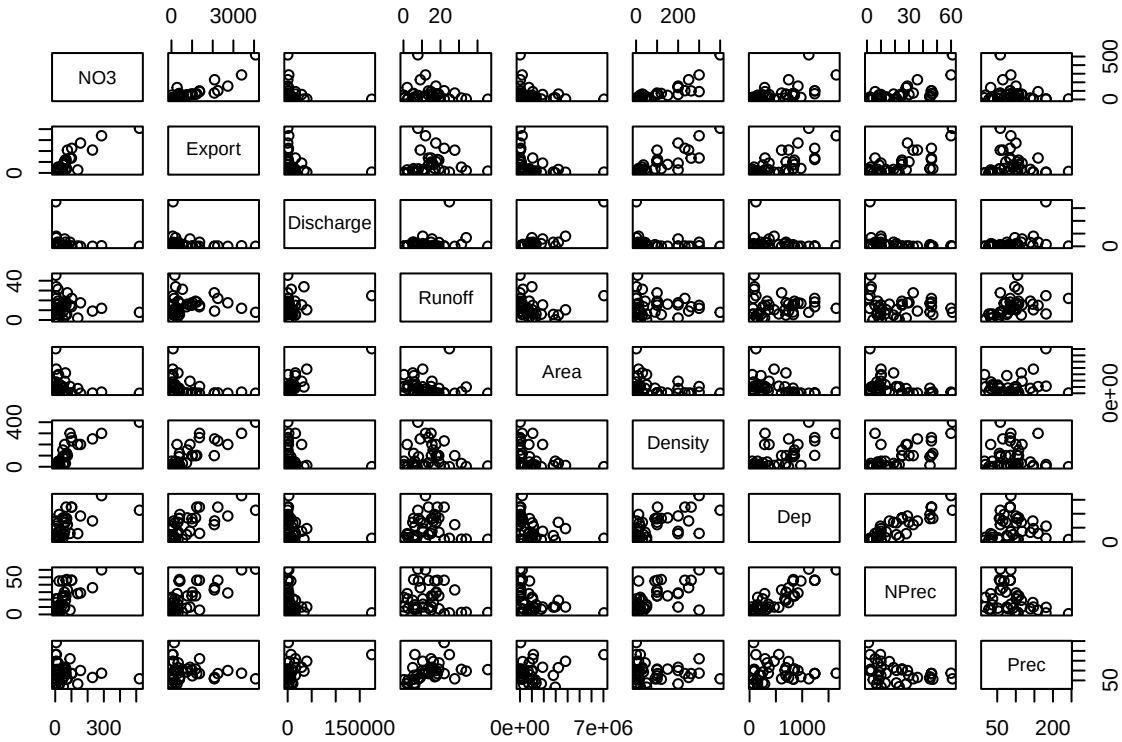
```
##       NO3             Export          Discharge             Runoff
##  Min.   :  1.10   Min.   :   4.40   Min.   :     7.29   Min.   : 0.20
##  1st Qu.:  9.30   1st Qu.:  76.03   1st Qu.:   392.00   1st Qu.: 6.10
##  Median : 29.50   Median : 314.30   Median :  2050.00   Median :11.85
##  Mean   : 62.32   Mean   : 731.71   Mean   : 10342.30   Mean   :13.24
##  3rd Qu.: 66.75   3rd Qu.:1008.30   3rd Qu.:  8125.00   3rd Qu.:17.65
##  Max.   :520.00   Max.   :4076.40   Max.   :175000.00   Max.   :45.60
##       Area            Density            Dep             NPrec
##  Min.   :    160   Min.   :  0.15   Min.   :  50.9   Min.   : 1.00
##  1st Qu.:  43828   1st Qu.: 10.00   1st Qu.: 141.8   1st Qu.: 7.65
##  Median : 517500   Median : 32.50   Median : 363.1   Median :14.85
##  Mean   : 937888   Mean   : 85.36   Mean   : 512.2   Mean   :20.79
##  3rd Qu.:1072250   3rd Qu.:115.50   3rd Qu.: 825.2   3rd Qu.:29.73
##  Max.   :7050000   Max.   :400.00   Max.   :1647.9   Max.   :61.00
##       Prec
```

```
##  Min.   : 15.70
##  1st Qu.: 57.98
##  Median : 85.45
##  Mean   : 89.84
##  3rd Qu.:107.10
##  Max.   :242.10
```

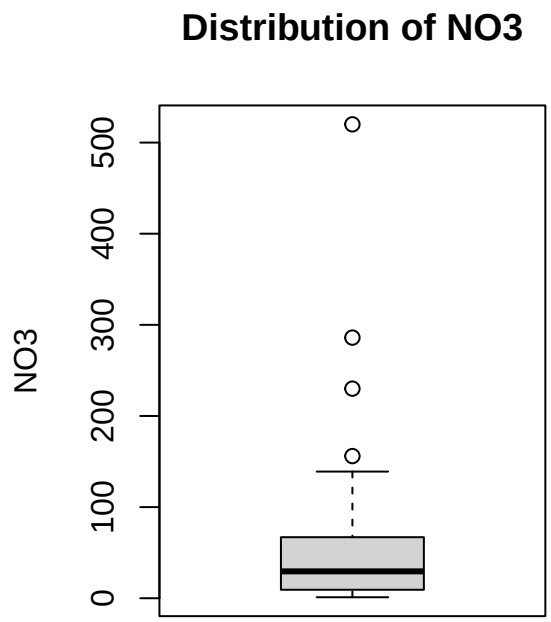
```
cor(rivernitrogen[, c("NO3","Export", "Discharge","Runoff","Area","Density","Dep","NPrec","Prec")])
```

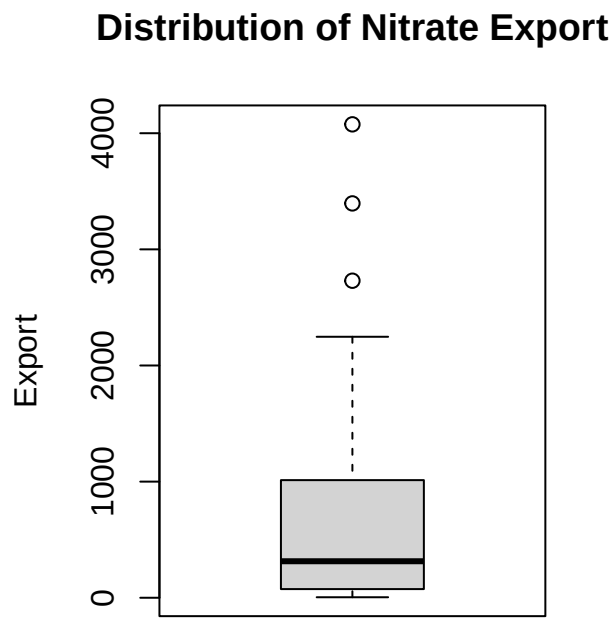
```
##                  NO3      Export  Discharge      Runoff       Area     Density
## NO3        1.0000000  0.87760374 -0.1740562 -0.10474564 -0.2733035  0.84100494
## Export     0.8776037  1.00000000 -0.1852446  0.15820802 -0.3748418  0.85438918
## Discharge -0.1740562 -0.18524457  1.0000000  0.20856498  0.8266886 -0.17507628
## Runoff    -0.1047456  0.15820802  0.2085650  1.00000000 -0.1611179 -0.01587029
## Area      -0.2733035 -0.37484182  0.8266886 -0.16111786  1.0000000 -0.30496113
## Density    0.8410049  0.85438918 -0.1750763 -0.01587029 -0.3049611  1.00000000
## Dep        0.5747647  0.70896977 -0.2267561  0.04663907 -0.3696825  0.64082072
## NPrec      0.6821405  0.69687082 -0.2942607 -0.11597582 -0.3982112  0.66893792
## Prec      -0.1663476 -0.04540047  0.3856870  0.45397595  0.1902521 -0.06956628
##                   Dep      NPrec        Prec
## NO3        0.57476471  0.6821405 -0.16634759
## Export     0.70896977  0.6968708 -0.04540047
## Discharge -0.22675609 -0.2942607  0.38568698
## Runoff     0.04663907 -0.1159758  0.45397595
## Area      -0.36968253 -0.3982112  0.19025208
## Density    0.64082072  0.6689379 -0.06956628
## Dep        1.00000000  0.8895985  0.01234455
## NPrec      0.88959851  1.0000000 -0.31829493
## Prec       0.01234455 -0.3182949  1.00000000
```

```
pairs(rivernitrogen[, c("NO3","Export", "Discharge","Runoff","Area","Density","Dep","NPrec","Prec")])
```

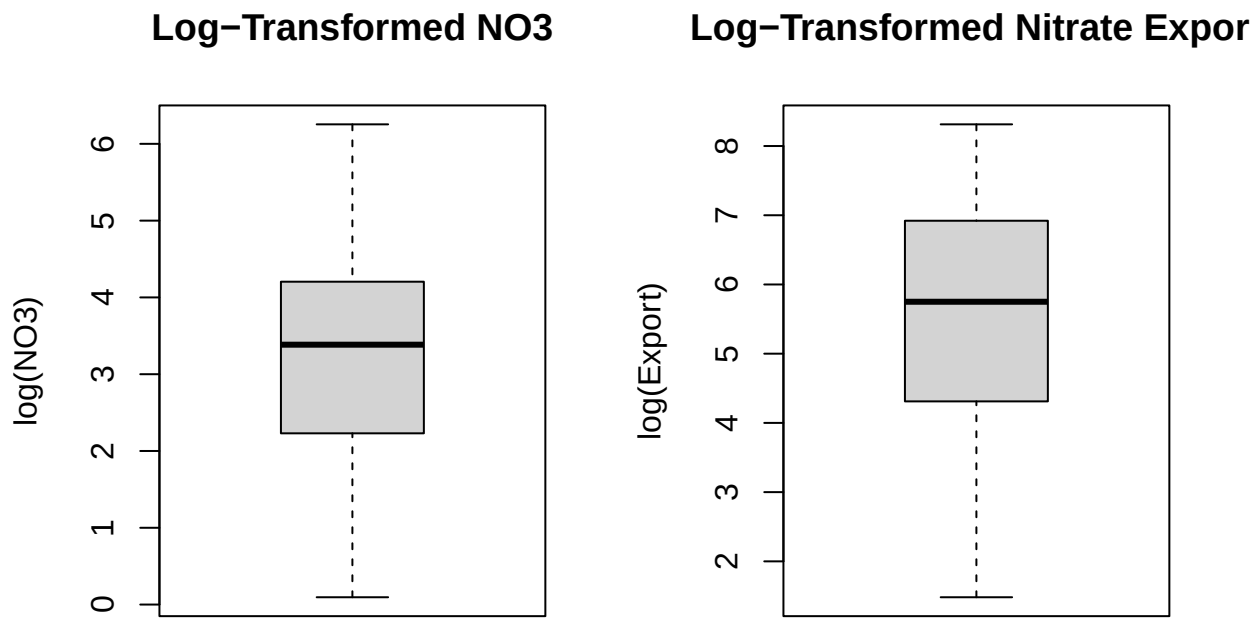



```
library(ggplot2)

par(mfrow = c(1, 2))

boxplot(rivernitrogen$NO3, main = "Distribution of NO3", ylab = "NO3")
boxplot(rivernitrogen$Export, main = "Distribution of Nitrate Export", ylab = "Export")
```

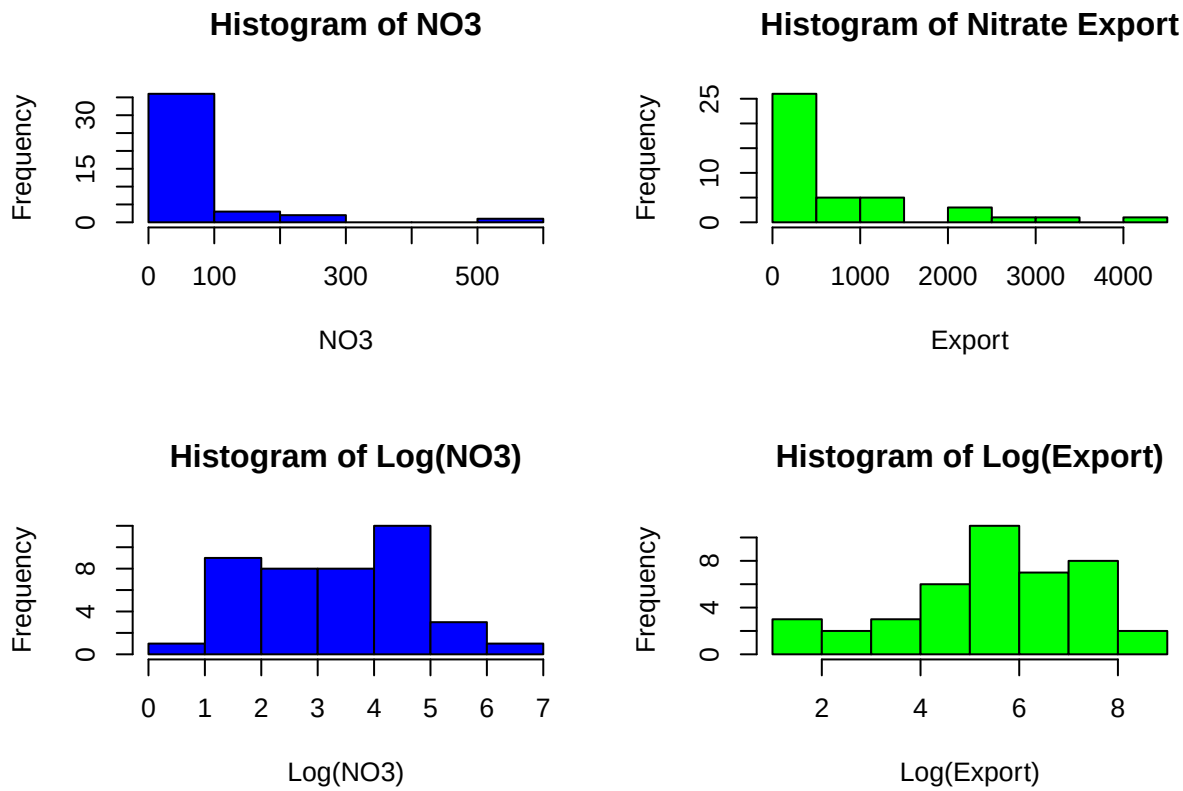





```
boxplot(log(rivernitrogen$NO3), main = "Log-Transformed NO3", ylab = "log(NO3)")
boxplot(log(rivernitrogen$Export), main = "Log-Transformed Nitrate Export", ylab = "log(Export)")
```




```
par(mfrow = c(2, 2))

hist(rivernitrogen$NO3, main = "Histogram of NO3", xlab = "NO3", col = "blue", border = "black")
hist(rivernitrogen$Export, main = "Histogram of Nitrate Export", xlab = "Export", col = "green", border

hist(log(rivernitrogen$NO3), main = "Histogram of Log(NO3)", xlab = "Log(NO3)", col = "blue", border =
hist(log(rivernitrogen$Export), main = "Histogram of Log(Export)", xlab = "Log(Export)", col = "green",
```



The histograms indicate that both **NO3** and **Nitrate Export** exhibit **right-skewed distributions**, suggesting non-normality. Applying a **log transformation** results in a more **symmetrical and approximately normal distribution**, improving statistical assumptions for analysis.

Thus, a **log transformation is appropriate** for normalizing these variables before modeling.

Statistical Modeling

```r
# Linear regression model for NO3
model_NO3 <- lm(NO3 ~ Discharge + Runoff + Prec + Area + Dep + NPrec + Density, data = rivernitrogen)
summary(model_NO3)
```

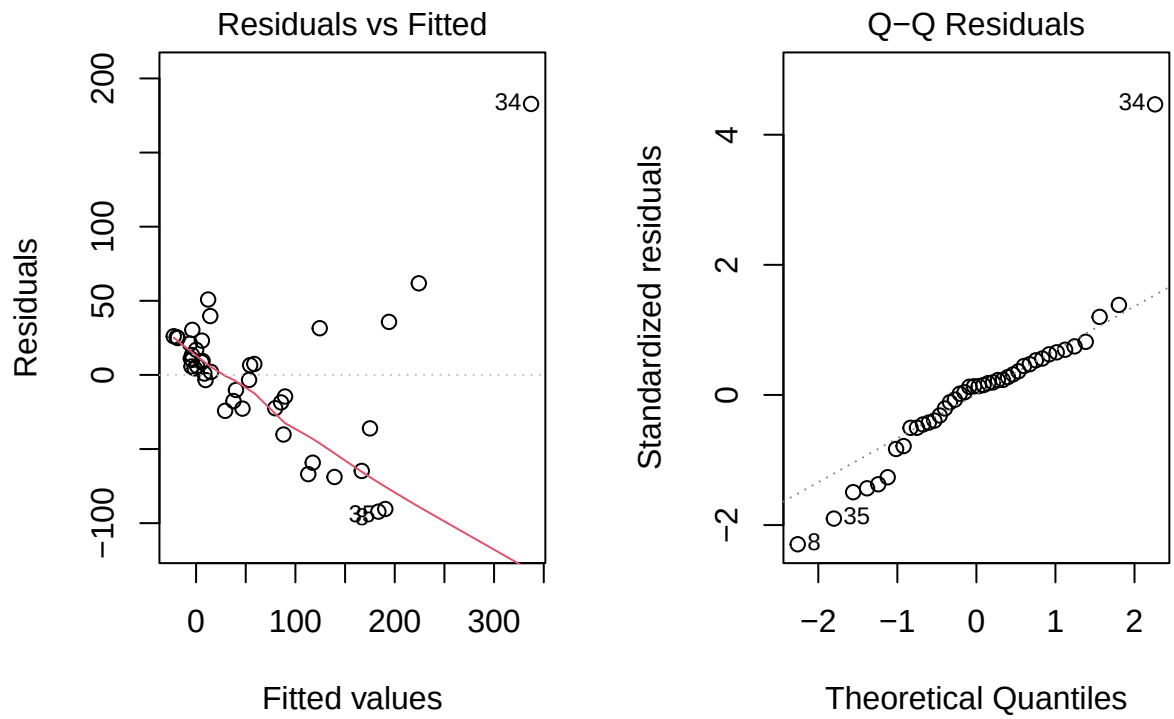
```
##
## Call:
## lm(formula = NO3 ~ Discharge + Runoff + Prec + Area + Dep + NPrec +
##     Density, data = rivernitrogen)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -92.222 -21.480   5.824  22.622 182.781
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.481e+01  3.763e+01  -0.659   0.5142
## Discharge    4.649e-05  7.041e-04   0.066   0.9477
```

```
## Runoff      -5.299e-01  1.158e+00  -0.457   0.6503
## Prec         2.035e-01  2.770e-01   0.735   0.4676
## Area        -6.812e-07  1.458e-05  -0.047   0.9630
## Dep         -9.425e-02  5.888e-02  -1.601   0.1187
## NPrec        3.351e+00  1.611e+00   2.080   0.0451 *
## Density      6.398e-01  1.051e-01   6.090 6.59e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 50.79 on 34 degrees of freedom
## Multiple R-squared:  0.7585, Adjusted R-squared:  0.7088
## F-statistic: 15.26 on 7 and 34 DF,  p-value: 7.736e-09
```

```r
anova(model_NO3)
```

```
## Analysis of Variance Table
##
## Response: NO3
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Discharge  1  11001   11001  4.2655 0.0465849 *
## Runoff     1   1778    1778  0.6896 0.4121080
## Prec       1   2705    2705  1.0488 0.3130114
## Area       1  42505   42505 16.4802 0.0002729 ***
## Dep        1  83264   83264 32.2832 2.231e-06 ***
## NPrec      1  38539   38539 14.9424 0.0004752 ***
## Density    1  95652   95652 37.0863 6.586e-07 ***
## Residuals 34  87692    2579
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
par(mfrow=c(1, 2))
plot(model_NO3, which=1)
plot(model_NO3, which=2)
```



The residuals vs. fitted plot indicates a non-random pattern, suggesting potential model misspecification or non-constant variance. The Q-Q plot shows deviations from normality, particularly in the upper tail, indicating that the residuals are not normally distributed. These issues suggest that model assumptions may not be fully satisfied, and log-transformations may be necessary.

```
# The log-transformed model for NO3
rivernitrogen$log_NO3 <- log(rivernitrogen$NO3)
model_log_NO3 <- lm(log_NO3 ~ Discharge + Runoff + Prec + Area + Dep + NPrec + Density, data = rivernit
summary(model_log_NO3)
```

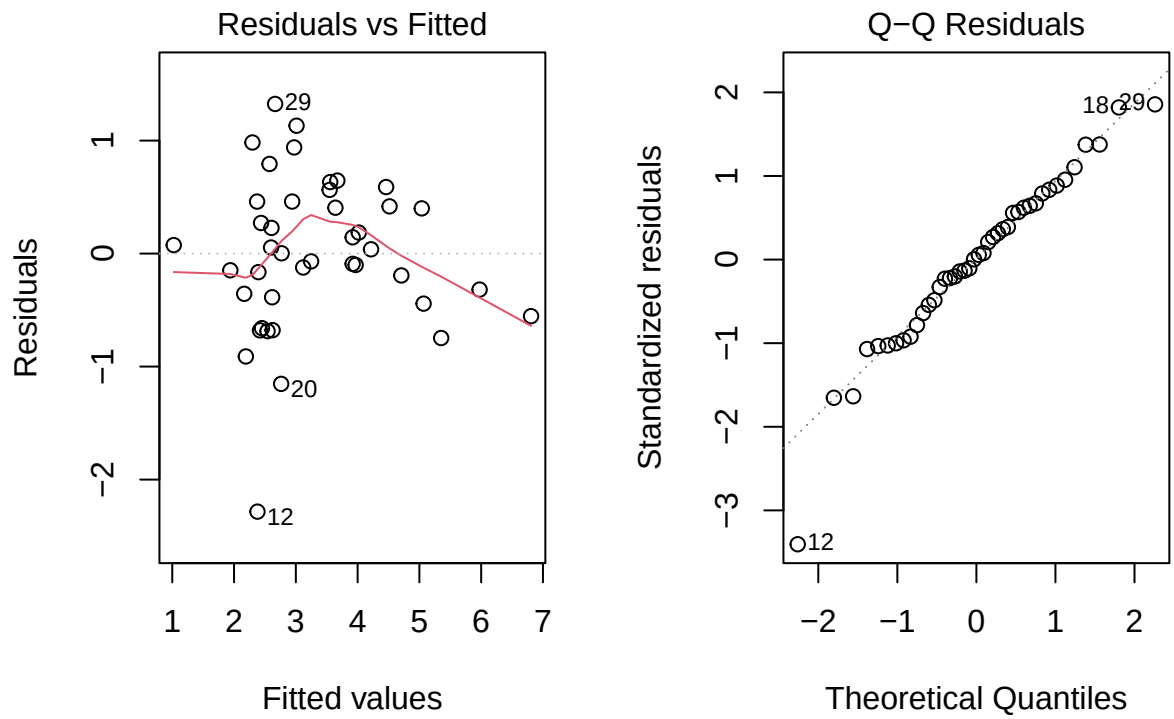
```
##
## Call:
## lm(formula = log_NO3 ~ Discharge + Runoff + Prec + Area + Dep +
##     NPrec + Density, data = rivernitrogen)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.28370 -0.37884  0.02017  0.44956  1.32416
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.247e+00  5.521e-01   4.070 0.000265 ***
## Discharge   -1.029e-05  1.033e-05  -0.996 0.326223
## Runoff      -3.470e-03  1.699e-02  -0.204 0.839442
## Prec        -1.350e-04  4.064e-03  -0.033 0.973699
## Area         8.929e-08  2.139e-07   0.417 0.679008
## Dep          6.478e-05  8.638e-04   0.075 0.940661
```

```
## NPrec        2.022e-02  2.364e-02   0.855 0.398323
## Density      8.223e-03  1.541e-03   5.335  6.3e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.745 on 34 degrees of freedom
## Multiple R-squared:  0.7519, Adjusted R-squared:  0.7009
## F-statistic: 14.72 on 7 and 34 DF,  p-value: 1.197e-08
```

```
anova(model_log_NO3)
```

```
## Analysis of Variance Table
##
## Response: log_NO3
##           Df  Sum Sq Mean Sq F value    Pr(>F)
## Discharge  1  8.3393  8.3393 15.0235 0.0004613 ***
## Runoff     1  0.2325  0.2325  0.4189 0.5218213
## Prec       1  0.3219  0.3219  0.5799 0.4516109
## Area       1  5.9650  5.9650 10.7461 0.0024139 **
## Dep        1 23.3998 23.3998 42.1554 1.988e-07 ***
## NPrec      1  3.1511  3.1511  5.6768 0.0229253 *
## Density    1 15.8001 15.8001 28.4643 6.304e-06 ***
## Residuals 34 18.8729  0.5551
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
par(mfrow=c(1, 2))
plot(model_log_NO3, which=1)
plot(model_log_NO3, which=2)
```



The log-transformed model for NO3 improves normality and variance homogeneity. The Q-Q plot indicates residuals align more closely with the normal distribution, reducing skewness. The residuals vs. fitted plot shows less heteroscedasticity compared to the untransformed model, though some curvature remains.

```
library(MASS)
model_NO3_step <- step(lm(log(NO3) ~ log(Density) + log(Dep) + log(NPrec) + log(Runoff) + log(Discharge
```

```
## Start:  AIC=-26.3
## log(NO3) ~ log(Density) + log(Dep) + log(NPrec) + log(Runoff) +
##     log(Discharge) + log(Area) + log(Prec)
##
##                  Df Sum of Sq    RSS      AIC
## - log(Runoff)     1    0.0007 15.343 -28.2955
## - log(Area)       1    0.0548 15.397 -28.1478
## - log(Dep)        1    0.0678 15.410 -28.1122
## - log(Prec)       1    0.0819 15.424 -28.0739
## - log(NPrec)      1    0.0960 15.438 -28.0354
## - log(Discharge)  1    0.1304 15.472 -27.9420
## <none>                        15.342 -26.2975
## - log(Density)    1   20.2759 35.618   7.0773
##
## Step:  AIC=-28.3
## log(NO3) ~ log(Density) + log(Dep) + log(NPrec) + log(Discharge) +
##     log(Area) + log(Prec)
##
##                  Df Sum of Sq    RSS      AIC
```

```
## - log(Dep)        1    0.0672 15.410 -30.1119
## - log(Prec)       1    0.0812 15.424 -30.0738
## - log(NPrec)      1    0.0954 15.438 -30.0352
## - log(Area)       1    0.4418 15.784 -29.1031
## - log(Discharge)  1    0.7384 16.081 -28.3214
## <none>                        15.343 -28.2955
## + log(Runoff)     1    0.0007 15.342 -26.2975
## - log(Density)    1   20.3254 35.668   5.1365
##
## Step:  AIC=-30.11
## log(NO3) ~ log(Density) + log(NPrec) + log(Discharge) + log(Area) +
##     log(Prec)
##
##                  Df Sum of Sq    RSS     AIC
## - log(Prec)       1    0.1328 15.543 -31.752
## - log(Area)       1    0.3819 15.792 -31.084
## - log(Discharge)  1    0.6712 16.081 -30.321
## <none>                        15.410 -30.112
## + log(Dep)        1    0.0672 15.343 -28.295
## + log(Runoff)     1    0.0001 15.410 -28.112
## - log(NPrec)      1    1.6783 17.088 -27.770
## - log(Density)    1   20.3352 35.745   3.227
##
## Step:  AIC=-31.75
## log(NO3) ~ log(Density) + log(NPrec) + log(Discharge) + log(Area)
##
##                  Df Sum of Sq    RSS     AIC
## - log(Area)       1    0.2567 15.799 -33.064
## - log(Discharge)  1    0.6211 16.164 -32.106
## <none>                        15.543 -31.752
## + log(Prec)       1    0.1328 15.410 -30.112
## + log(Dep)        1    0.1188 15.424 -30.074
## + log(Runoff)     1    0.0058 15.537 -29.767
## - log(NPrec)      1    1.5682 17.111 -29.714
## - log(Density)    1   22.8513 38.394   4.230
##
## Step:  AIC=-33.06
## log(NO3) ~ log(Density) + log(NPrec) + log(Discharge)
##
##                  Df Sum of Sq    RSS     AIC
## - log(Discharge)  1    0.4438 16.243 -33.900
## <none>                        15.799 -33.064
## + log(Area)       1    0.2567 15.543 -31.752
## + log(Runoff)     1    0.2180 15.581 -31.647
## + log(Dep)        1    0.0084 15.791 -31.086
## + log(Prec)       1    0.0076 15.792 -31.084
## - log(NPrec)      1    1.8006 17.600 -30.531
## - log(Density)    1   22.8702 38.669   2.530
##
## Step:  AIC=-33.9
## log(NO3) ~ log(Density) + log(NPrec)
##
##                  Df Sum of Sq    RSS     AIC
## <none>                        16.243 -33.900
```

```
## + log(Discharge)  1    0.4438 15.799 -33.064
## + log(Runoff)     1    0.2561 15.987 -32.568
## + log(Area)       1    0.0795 16.164 -32.106
## + log(Prec)       1    0.0529 16.190 -32.037
## + log(Dep)        1    0.0523 16.191 -32.035
## - log(NPrec)      1    2.2988 18.542 -30.341
## - log(Density)    1   24.0758 40.319   2.284
```

```
model_Export_step <- step(lm(log(Export) ~ log(Density) + log(Dep) + log(NPrec) + log(Runoff) + log(Dis
```

```
## Start:  AIC=-26.91
## log(Export) ~ log(Density) + log(Dep) + log(NPrec) + log(Runoff) +
##     log(Discharge) + log(Area) + log(Prec)
##
##                  Df Sum of Sq    RSS      AIC
## - log(Area)       1    0.0507 15.169 -28.7732
## - log(Discharge)  1    0.1239 15.242 -28.5710
## - log(Dep)        1    0.1513 15.270 -28.4956
## - log(Prec)       1    0.1708 15.289 -28.4419
## - log(NPrec)      1    0.1920 15.310 -28.3839
## <none>                        15.118 -26.9138
## - log(Runoff)     1    3.7905 18.909 -19.5175
## - log(Density)    1   20.4643 35.583   7.0359
##
## Step:  AIC=-28.77
## log(Export) ~ log(Density) + log(Dep) + log(NPrec) + log(Runoff) +
##     log(Discharge) + log(Prec)
##
##                  Df Sum of Sq    RSS      AIC
## - log(Dep)        1    0.1440 15.313 -30.3764
## - log(Prec)       1    0.1627 15.332 -30.3252
## - log(NPrec)      1    0.1827 15.352 -30.2703
## - log(Discharge)  1    0.5444 15.714 -29.2921
## <none>                        15.169 -28.7732
## + log(Area)       1    0.0507 15.118 -26.9138
## - log(Runoff)     1   19.8956 35.065   4.4201
## - log(Density)    1   20.6284 35.798   5.2888
##
## Step:  AIC=-30.38
## log(Export) ~ log(Density) + log(NPrec) + log(Runoff) + log(Discharge) +
##     log(Prec)
##
##                  Df Sum of Sq    RSS     AIC
## - log(Prec)       1    0.1127 15.426 -32.069
## - log(Discharge)  1    0.4541 15.767 -31.149
## <none>                        15.313 -30.376
## + log(Dep)        1    0.1440 15.169 -28.773
## + log(Area)       1    0.0434 15.270 -28.496
## - log(NPrec)      1    1.6042 16.917 -28.192
## - log(Density)    1   20.5195 35.833   3.330
## - log(Runoff)     1   22.4638 37.777   5.549
##
## Step:  AIC=-32.07
## log(Export) ~ log(Density) + log(NPrec) + log(Runoff) + log(Discharge)
```

```
##
##                   Df Sum of Sq    RSS     AIC
## - log(Discharge)  1     0.372 15.797 -33.068
## <none>                        15.426 -32.069
## + log(Prec)       1     0.113 15.313 -30.376
## + log(Dep)        1     0.094 15.332 -30.325
## + log(Area)       1     0.039 15.387 -30.174
## - log(NPrec)      1     1.506 16.931 -30.157
## - log(Density)    1    22.649 38.075   3.879
## - log(Runoff)     1    50.278 65.704  26.794
##
## Step:  AIC=-33.07
## log(Export) ~ log(Density) + log(NPrec) + log(Runoff)
##
##                   Df Sum of Sq    RSS     AIC
## <none>                        15.797 -33.068
## + log(Discharge)  1     0.372 15.426 -32.069
## + log(Area)       1     0.322 15.475 -31.934
## + log(Prec)       1     0.030 15.767 -31.149
## + log(Dep)        1     0.024 15.773 -31.134
## - log(NPrec)      1     1.882 17.679 -30.341
## - log(Density)    1    23.905 39.702   3.637
## - log(Runoff)     1    49.940 65.737  24.816
```

```r
# Stepwise Model Selection for NO3
model_NO3_forward <- step(lm(log(NO3) ~ 1, data = rivernitrogen),
                          scope = ~ log(Density) + log(Dep) + log(NPrec) + log(Runoff) + log(Discharge)
                          direction = "forward")
```

```
## Start:  AIC=26.95
## log(NO3) ~ 1
##
##                   Df Sum of Sq    RSS      AIC
## + log(Density)    1    57.541 18.542 -30.3407
## + log(NPrec)      1    35.764 40.319   2.2844
## + log(Dep)        1    33.016 43.067   5.0536
## + log(Discharge)  1    11.066 65.016  22.3527
## + log(Area)       1     9.270 66.813  23.4974
## <none>                        76.083  26.9544
## + log(Prec)       1     0.304 75.779  28.7862
## + log(Runoff)     1     0.018 76.064  28.9443
##
## Step:  AIC=-30.34
## log(NO3) ~ log(Density)
##
##                   Df Sum of Sq    RSS     AIC
## + log(NPrec)      1   2.29879 16.243 -33.900
## + log(Discharge)  1   0.94208 17.600 -30.531
## + log(Dep)        1   0.89175 17.650 -30.411
## <none>                        18.542 -30.341
## + log(Prec)       1   0.70550 17.836 -29.970
## + log(Runoff)     1   0.66289 17.879 -29.870
## + log(Area)       1   0.18257 18.359 -28.756
##
```

```
## Step:  AIC=-33.9
## log(NO3) ~ log(Density) + log(NPrec)
##
##                  Df Sum of Sq    RSS     AIC
## <none>                        16.243 -33.900
## + log(Discharge)  1   0.44385 15.799 -33.064
## + log(Runoff)     1   0.25615 15.987 -32.568
## + log(Area)       1   0.07948 16.164 -32.106
## + log(Prec)       1   0.05294 16.190 -32.037
## + log(Dep)        1   0.05225 16.191 -32.035
```

```
model_NO3_backward <- step(lm(log(NO3) ~ log(Density) + log(Dep) + log(NPrec) + log(Runoff) + log(Discha
```

```
## Start:  AIC=-26.3
## log(NO3) ~ log(Density) + log(Dep) + log(NPrec) + log(Runoff) +
##     log(Discharge) + log(Area) + log(Prec)
##
##                  Df Sum of Sq    RSS      AIC
## - log(Runoff)     1    0.0007 15.343 -28.2955
## - log(Area)       1    0.0548 15.397 -28.1478
## - log(Dep)        1    0.0678 15.410 -28.1122
## - log(Prec)       1    0.0819 15.424 -28.0739
## - log(NPrec)      1    0.0960 15.438 -28.0354
## - log(Discharge)  1    0.1304 15.472 -27.9420
## <none>                        15.342 -26.2975
## - log(Density)    1   20.2759 35.618   7.0773
##
## Step:  AIC=-28.3
## log(NO3) ~ log(Density) + log(Dep) + log(NPrec) + log(Discharge) +
##     log(Area) + log(Prec)
##
##                  Df Sum of Sq    RSS      AIC
## - log(Dep)        1    0.0672 15.410 -30.1119
## - log(Prec)       1    0.0812 15.424 -30.0738
## - log(NPrec)      1    0.0954 15.438 -30.0352
## - log(Area)       1    0.4418 15.784 -29.1031
## - log(Discharge)  1    0.7384 16.081 -28.3214
## <none>                        15.343 -28.2955
## - log(Density)    1   20.3254 35.668   5.1365
##
## Step:  AIC=-30.11
## log(NO3) ~ log(Density) + log(NPrec) + log(Discharge) + log(Area) +
##     log(Prec)
##
##                  Df Sum of Sq    RSS     AIC
## - log(Prec)       1    0.1328 15.543 -31.752
## - log(Area)       1    0.3819 15.792 -31.084
## - log(Discharge)  1    0.6712 16.081 -30.321
## <none>                        15.410 -30.112
## - log(NPrec)      1    1.6783 17.088 -27.770
## - log(Density)    1   20.3352 35.745   3.227
##
## Step:  AIC=-31.75
## log(NO3) ~ log(Density) + log(NPrec) + log(Discharge) + log(Area)
```

```
##
##                  Df Sum of Sq    RSS     AIC
## - log(Area)       1    0.2567 15.799 -33.064
## - log(Discharge)  1    0.6211 16.164 -32.106
## <none>                        15.543 -31.752
## - log(NPrec)      1    1.5682 17.111 -29.714
## - log(Density)    1   22.8513 38.394   4.230
##
## Step:  AIC=-33.06
## log(NO3) ~ log(Density) + log(NPrec) + log(Discharge)
##
##                  Df Sum of Sq    RSS     AIC
## - log(Discharge)  1    0.4438 16.243 -33.900
## <none>                        15.799 -33.064
## - log(NPrec)      1    1.8006 17.600 -30.531
## - log(Density)    1   22.8702 38.669   2.530
##
## Step:  AIC=-33.9
## log(NO3) ~ log(Density) + log(NPrec)
##
##                Df Sum of Sq    RSS     AIC
## <none>                      16.243 -33.900
## - log(NPrec)    1    2.2988 18.542 -30.341
## - log(Density)  1   24.0758 40.319   2.284
```

```r
# Linear Model for Nitrate Export
model_Export <- lm(Export ~ Discharge + Runoff + Prec + Area + Dep + NPrec + Density, data = rivernitrog
summary(model_Export)
```

```
##
## Call:
## lm(formula = Export ~ Discharge + Runoff + Prec + Area + Dep +
##     NPrec + Density, data = rivernitrogen)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -963.92 -161.74   33.88  166.55 1023.11
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.519e+02  3.483e+02  -0.436    0.666
## Discharge    1.268e-03  6.516e-03   0.195    0.847
## Runoff       1.788e+01  1.072e+01   1.668    0.105
## Prec        -1.848e+00  2.563e+00  -0.721    0.476
## Area        -4.236e-05  1.349e-04  -0.314    0.755
## Dep          6.216e-01  5.449e-01   1.141    0.262
## NPrec       -8.843e-01  1.491e+01  -0.059    0.953
## Density      6.321e+00  9.723e-01   6.501 1.94e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 470 on 34 degrees of freedom
## Multiple R-squared:  0.8061, Adjusted R-squared:  0.7661
## F-statistic: 20.19 on 7 and 34 DF,  p-value: 2.144e-10
```

```r
anova(model_Export)
```

```
## Analysis of Variance Table
##
## Response: Export
##           Df   Sum Sq  Mean Sq F value    Pr(>F)
## Discharge  1  1328688  1328688  6.0157   0.01946 *
## Runoff     1  1568516  1568516  7.1016   0.01169 *
## Prec       1   141803   141803  0.6420   0.42854
## Area       1  4522272  4522272 20.4749 7.032e-05 ***
## Dep        1 13631280 13631280 61.7166 3.804e-09 ***
## NPrec      1   683519   683519  3.0947   0.08755 .
## Density    1  9334068  9334068 42.2606 1.941e-07 ***
## Residuals 34  7509549   220869
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Log-Transformed Model for Nitrate Export
model_log_Export <- lm(log(Export) ~ Discharge + Runoff + Prec + Area + Dep + NPrec + Density, data = ri
summary(model_log_Export)
```

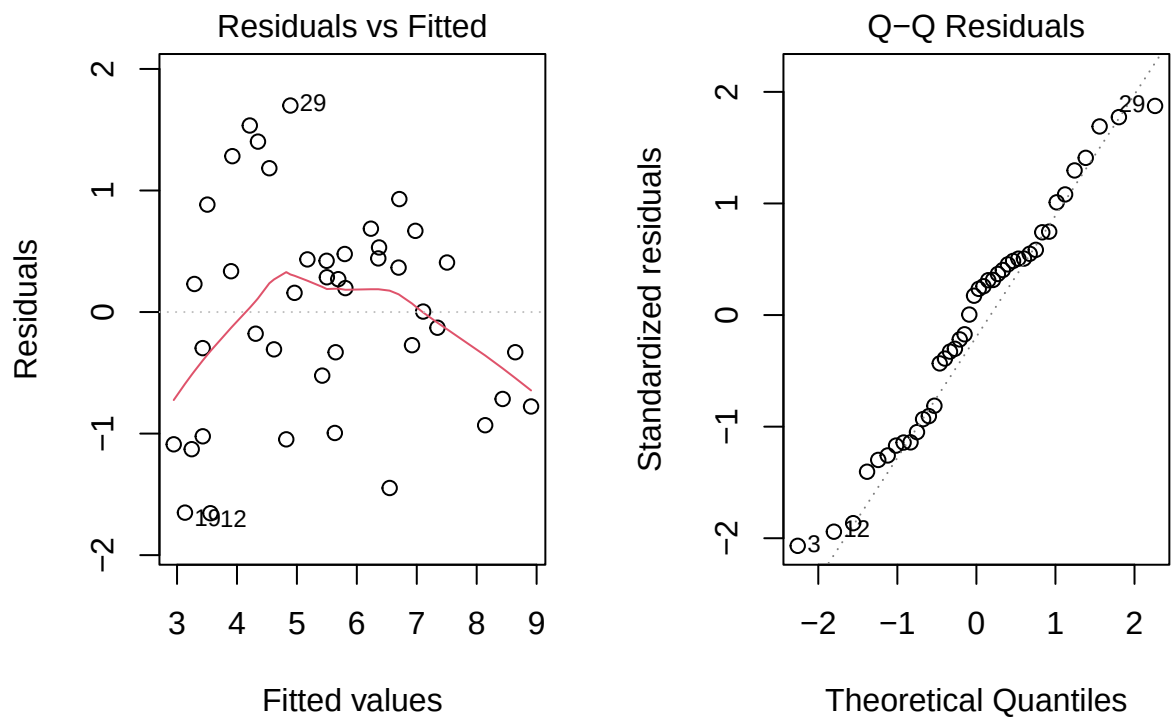
```
##
## Call:
## lm(formula = log(Export) ~ Discharge + Runoff + Prec + Area +
##     Dep + NPrec + Density, data = rivernitrogen)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.6551 -0.6672  0.1774  0.4683  1.6988
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.053e+00  7.018e-01   4.350 0.000117 ***
## Discharge    4.130e-06  1.313e-05   0.315 0.755034
## Runoff       6.474e-02  2.160e-02   2.997 0.005064 **
## Prec         3.437e-03  5.166e-03   0.665 0.510293
## Area        -2.223e-07  2.719e-07  -0.818 0.419280
## Dep          1.476e-03  1.098e-03   1.344 0.187878
## NPrec       -2.375e-03  3.005e-02  -0.079 0.937454
## Density      8.430e-03  1.959e-03   4.303 0.000135 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9471 on 34 degrees of freedom
## Multiple R-squared:  0.7804, Adjusted R-squared:  0.7351
## F-statistic: 17.26 on 7 and 34 DF,  p-value: 1.649e-09
```

```r
anova(model_log_Export)
```

```
## Analysis of Variance Table
##
## Response: log(Export)
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Discharge  1  2.051   2.051  2.2871 0.1396933
## Runoff     1 30.326  30.326 33.8090 1.500e-06 ***
## Prec       1  1.340   1.340  1.4938 0.2300278
## Area       1 16.080  16.080 17.9271 0.0001648 ***
## Dep        1 40.816  40.816 45.5038 9.421e-08 ***
## NPrec      1  1.131   1.131  1.2611 0.2693174
## Density    1 16.605  16.605 18.5119 0.0001350 ***
## Residuals 34 30.497   0.897
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
par(mfrow=c(1, 2))
plot(model_log_Export, which=1)
plot(model_log_Export, which=2)
```




```
# Stepwise Model Selection for Nitrate Export
model_Export_forward <- step(lm(log(Export) ~ 1, data = rivernitrogen), scope = ~ log(Density) + log(De
```

```
## Start:  AIC=52.22
## log(Export) ~ 1
##
##                  Df Sum of Sq     RSS    AIC
## + log(Density)    1    73.095  65.751 22.825
## + log(Dep)        1    67.258  71.588 26.397
```

```
## + log(Runoff)     1    63.083  75.763 28.778
## + log(Area)       1    43.011  95.835 38.648
## + log(NPrec)      1    28.365 110.481 44.621
## + log(Prec)       1    25.253 113.593 45.788
## + log(Discharge)  1     8.168 130.679 51.673
## <none>                        138.846 52.219
##
## Step:  AIC=22.82
## log(Export) ~ log(Density)
##
##                  Df Sum of Sq    RSS      AIC
## + log(Runoff)     1    48.072 17.679 -30.3411
## + log(Prec)       1    22.128 43.624   7.5932
## + log(Area)       1    14.573 51.178  14.3010
## + log(Dep)        1    11.374 54.378  16.8480
## <none>                        65.751  22.8249
## + log(Discharge)  1     0.024 65.728  24.8097
## + log(NPrec)      1     0.014 65.737  24.8157
##
## Step:  AIC=-30.34
## log(Export) ~ log(Density) + log(Runoff)
##
##                  Df Sum of Sq    RSS     AIC
## + log(NPrec)      1   1.88202 15.797 -33.068
## + log(Dep)        1   1.57112 16.108 -32.250
## <none>                        17.680 -30.341
## + log(Discharge)  1   0.74801 16.931 -30.157
## + log(Area)       1   0.72365 16.956 -30.096
## + log(Prec)       1   0.13886 17.541 -28.672
##
## Step:  AIC=-33.07
## log(Export) ~ log(Density) + log(Runoff) + log(NPrec)
##
##                  Df Sum of Sq    RSS     AIC
## <none>                        15.797 -33.068
## + log(Discharge)  1   0.37171 15.426 -32.069
## + log(Area)       1   0.32240 15.475 -31.934
## + log(Prec)       1   0.03029 15.767 -31.149
## + log(Dep)        1   0.02448 15.773 -31.134
```

```
model_Export_backward <- step(lm(log(Export) ~ log(Density) + log(Dep) + log(NPrec) + log(Runoff) + log
```

```
## Start:  AIC=-26.91
## log(Export) ~ log(Density) + log(Dep) + log(NPrec) + log(Runoff) +
##     log(Discharge) + log(Area) + log(Prec)
##
##                  Df Sum of Sq    RSS      AIC
## - log(Area)       1    0.0507 15.169 -28.7732
## - log(Discharge)  1    0.1239 15.242 -28.5710
## - log(Dep)        1    0.1513 15.270 -28.4956
## - log(Prec)       1    0.1708 15.289 -28.4419
## - log(NPrec)      1    0.1920 15.310 -28.3839
## <none>                        15.118 -26.9138
## - log(Runoff)     1    3.7905 18.909 -19.5175
```

```
## - log(Density)    1   20.4643 35.583   7.0359
##
## Step:  AIC=-28.77
## log(Export) ~ log(Density) + log(Dep) + log(NPrec) + log(Runoff) +
##     log(Discharge) + log(Prec)
##
##                  Df Sum of Sq    RSS      AIC
## - log(Dep)        1    0.1440 15.313 -30.3764
## - log(Prec)       1    0.1627 15.332 -30.3252
## - log(NPrec)      1    0.1827 15.352 -30.2703
## - log(Discharge)  1    0.5444 15.714 -29.2921
## <none>                        15.169 -28.7732
## - log(Runoff)     1   19.8956 35.065   4.4201
## - log(Density)    1   20.6284 35.798   5.2888
##
## Step:  AIC=-30.38
## log(Export) ~ log(Density) + log(NPrec) + log(Runoff) + log(Discharge) +
##     log(Prec)
##
##                  Df Sum of Sq    RSS     AIC
## - log(Prec)       1    0.1127 15.426 -32.069
## - log(Discharge)  1    0.4541 15.767 -31.149
## <none>                        15.313 -30.376
## - log(NPrec)      1    1.6042 16.917 -28.192
## - log(Density)    1   20.5195 35.833   3.330
## - log(Runoff)     1   22.4638 37.777   5.549
##
## Step:  AIC=-32.07
## log(Export) ~ log(Density) + log(NPrec) + log(Runoff) + log(Discharge)
##
##                  Df Sum of Sq    RSS     AIC
## - log(Discharge)  1     0.372 15.797 -33.068
## <none>                        15.426 -32.069
## - log(NPrec)      1     1.506 16.931 -30.157
## - log(Density)    1    22.649 38.075   3.879
## - log(Runoff)     1    50.278 65.704  26.794
##
## Step:  AIC=-33.07
## log(Export) ~ log(Density) + log(NPrec) + log(Runoff)
##
##                 Df Sum of Sq    RSS     AIC
## <none>                       15.797 -33.068
## - log(NPrec)    1     1.882 17.679 -30.341
## - log(Density)  1    23.905 39.702   3.637
## - log(Runoff)   1    49.940 65.737  24.816
```

Final Model Selection

```r
# Model for NO3
final_model_NO3 <- lm(log(NO3) ~ log(Density) + log(NPrec) + log(Discharge), data = rivernitrogen)
summary(final_model_NO3)
```

```
##
## Call:
## lm(formula = log(NO3) ~ log(Density) + log(NPrec) + log(Discharge),
##     data = rivernitrogen)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.33536 -0.35169 -0.08613  0.46772  1.06131
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     1.55572    0.54678   2.845  0.00711 **
## log(Density)    0.46320    0.06245   7.417 6.76e-09 ***
## log(NPrec)      0.26492    0.12730   2.081  0.04422 *
## log(Discharge) -0.05222    0.05054  -1.033  0.30803
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6448 on 38 degrees of freedom
## Multiple R-squared:  0.7923, Adjusted R-squared:  0.7759
## F-statistic: 48.33 on 3 and 38 DF,  p-value: 4.811e-13
```

```r
anova(final_model_NO3)
```

```
## Analysis of Variance Table
##
## Response: log(NO3)
##                Df Sum Sq Mean Sq  F value    Pr(>F)
## log(Density)    1 57.541  57.541 138.3954 3.101e-14 ***
## log(NPrec)      1  2.299   2.299   5.5290   0.02399 *
## log(Discharge)  1  0.444   0.444   1.0675   0.30803
## Residuals      38 15.799   0.416
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Model for Nitrate Export
final_model_Export <- lm(log(Export) ~ log(Density) + log(Runoff) + log(NPrec), data = rivernitrogen)
summary(final_model_Export)
```

```
##
## Call:
## lm(formula = log(Export) ~ log(Density) + log(Runoff) + log(NPrec),
##     data = rivernitrogen)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.20936 -0.39241 -0.08637  0.52508  1.16561
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.20645    0.34191   3.529  0.00111 **
## log(Density)  0.48260    0.06364   7.583 4.06e-09 ***
## log(Runoff)   0.94249    0.08599  10.960 2.51e-13 ***
```

```
## log(NPrec)    0.27085    0.12730   2.128  0.03991 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6448 on 38 degrees of freedom
## Multiple R-squared:  0.8862, Adjusted R-squared:  0.8772
## F-statistic: 98.66 on 3 and 38 DF,  p-value: < 2.2e-16
```

```
anova(final_model_Export)
```

```
## Analysis of Variance Table
##
## Response: log(Export)
##              Df Sum Sq Mean Sq  F value    Pr(>F)
## log(Density)  1 73.095  73.095 175.8260 7.836e-16 ***
## log(Runoff)   1 48.072  48.072 115.6347 4.365e-13 ***
## log(NPrec)    1  1.882   1.882   4.5271   0.03991 *
## Residuals    38 15.797   0.416
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusion

The final models for **NO3** and **Nitrate Export** were selected based on stepwise regression, optimizing AIC.

- For **NO3**, the significant predictors are **Density** ($p < 0.001$) and **NPrec** ($p < 0.05$), suggesting that human population density and nitrate precipitation are key factors influencing nitrate concentration in rivers. **Discharge** was not significant ($p > 0.05$).
- For **Nitrate Export**, significant predictors include **Density** ($p < 0.001$), **Runoff** ($p < 0.001$), and **NPrec** ($p < 0.05$), indicating that human population density, runoff, and nitrate precipitation drive nitrate export in rivers.

Both models exhibit strong explanatory power, with R^2 **values of 0.776 and 0.877**, respectively, suggesting a good fit. The results indicate that human activity, particularly population density, and hydrological factors (runoff and nitrate precipitation) play crucial roles in riverine nitrate pollution.